

Equivalent Circuit Parameterization from Electrochemical Impedance Spectroscopy Data for Accurate Battery Degradation Prediction using Convolution Neural Network

Latha Anekal

Dept. of Electrical, Computer and
Software Engineering
Faculty of Engineering and Applied
Science

Ontario Tech University
Oshawa, Canada

latha.anekal@ontariotechu.net

Chandan Chetri

Dept. of Electrical, Computer and
Software Engineering
Faculty of Engineering and Applied
Science

Ontario Tech University
Oshawa, Canada

chandan.chetri@ontariotechu.net

Meaghan Charest-Finn

Dept. of Automotive and Mechatronics
Engineering
Faculty of Engineering and Applied
Science

Ontario Tech University
Oshawa, Canada

meaghan.charest-finn@ontariotechu.ca

Sheldon Williamson

Dept. of Electrical, Computer and
Software Engineering
Faculty of Engineering and Applied
Science

Ontario Tech University
Oshawa, Canada

sheldon.williamson@ontariotechu.ca

Abstract—This paper presents a deep learning-based framework for the automated extraction and classification of equivalent circuit models (ECMs) from electrochemical impedance spectroscopy (EIS) data, aiming to enhance lithium-ion battery (LIB) diagnostics. EIS provides detailed insights into internal battery mechanisms, such as charge transfer, diffusion, and double-layer capacitance, by analyzing frequency-dependent impedance responses. Traditional interpretation methods are labour-intensive and limited in scalability. To address this, a one-dimensional convolutional neural network (1D-CNN) is employed to classify EIS spectra into four distinct ECM classes using features derived from real and synthetic datasets. The model architecture incorporates hierarchical convolutional layers, dropout, batch normalization, and global average pooling, achieving a classification accuracy of 95.65% on the test set. The predictions align closely with Nyquist plot characteristics of each ECM, validating the interpretability and robustness of the model.

Keywords— *Convolution neural network, electrochemical impedance spectroscopy, lithium-ion batteries, state of health.*

I. INTRODUCTION

Lithium-ion Batteries (LIBs) consist of two main electrodes, an anode and a cathode. These are divided separated by a porous separator and immersed in an electrolyte solution [1]. The cathode is typically made of lithium metal oxides and defines the battery chemistry. Common cathode materials include lithium manganese oxide (LMO), lithium cobalt oxide (LCO), lithium nickel manganese cobalt oxide (NMC), lithium iron phosphate (LFP), and lithium nickel cobalt aluminum oxide (NCA) [2]. Effective LIB performance requires electrode materials with open crystal structures that allow lithium intercalation and de-intercalation. LCO, NMC, and NCA have layered structures that facilitate ion movement [3], [4]. LMO adopts a 3D cubic spinel structure, enhancing ion transport and high-rate performance [5]. LFP features an

olivine (polyanion) structure, providing high thermal stability and safety, though it has lower conductivity and energy density [6].

Advancing battery technology is a cornerstone of sustainable engineering, particularly in the context of clean energy transitions and the growing demand for electric mobility and renewable energy integration [7], [8]. To meet these demands, intelligent battery management strategies are essential. One promising avenue lies in the application of machine learning (ML) techniques for improved battery health diagnostics and performance prediction [9], [10]. ML algorithms offer powerful tools to capture nonlinear and high-dimensional patterns in battery data, enabling the development of more adaptive and accurate Battery Management Systems (BMS) [11], [12]. A foundational step in this process is the accurate and efficient estimation of a State of Health (SOH) of the battery, which is a critical parameter that quantifies the aging and remaining useful life of a battery [13], [14]. Understanding the SOH of the battery enables better decision-making in battery operation, maintenance, and replacement, ultimately enhancing the reliability and safety of battery-powered systems. In recent years, rapid advancements in technology have led to the successful application of deep neural networks (DNNs) and various ML architectures across a range of scientific disciplines, including electrochemistry [15], [16], physics [17], [18] and astronomy [19]. Despite the growing adoption of ML in many scientific fields, its application in electrochemical systems remains relatively limited. In this work, we introduce a machine learning-based method using DNNs for the automatic classification of equivalent circuits from EIS data.

The paper is divided into the following sections: The details of battery modeling is discussed in Section II and the experimental setup is discussed in Section III. The DNNs for

EIS data extraction are discussed in Section IV, followed by the DNNs predicted equivalent circuit models (ECMs) and Nyquist plots in Section V. The DNN training and evaluation of EIS model results are discussed in Section VI. Finally, the conclusions are discussed in Section VII.

II. DETERMINISTIC BATTERY MODELING

A. Battery Kinetics

Battery kinetics refers to the internal processes that govern the movement of lithium ions (Li^+) and electrons during the charge and discharge cycles of LIBs. The primary electrochemical processes for these include charge transfer, double-layer capacitance, and diffusion [20]. The charge transfer process, shown in Fig. 1a, involves the movement of ions and electrons [10]. The activation energy required for lithium ions to traverse the electrode-electrolyte interface serves as a measure of the stability of the charge transfer process. When the SOH of a cell is greater than 80%, this mechanism remains stable, but when SOH drops below 80%, activation energy can decrease from 0.55 eV to 0.16 eV, increasing the risk of thermal runaway [21]. Another key process is the formation of the electrochemical double layer at the electrode-electrolyte interface, as shown in Fig. 1b, which behaves like a capacitor. The double-layer capacitance arises due to dielectric polarization, ion displacement, and ion/molecule adsorption [22]. The structure and behavior of the double layer depend on electrolyte composition and electrode potential, and play a significant role in lithium-ion storage and interfacial charge dynamics [23].

Diffusion of lithium ions within LIBs is another essential kinetic process, occurring in the electrolyte, electrodes, and interfacial layers, as shown in Fig. 1c. This influences the reaction rate of electrode materials and, consequently, the power performance of the battery [24]. Two concurrent behaviors dominate: (i) Li^+ insertion and extraction, involving movement into and out of the electrodes, and (ii) electron transfer, associated with redox reactions in the active material [25]. During these processes, variations in lithium concentration across the electrode generate gradients that drive diffusion. Additionally, the SEI layer forms spontaneously at the negative electrode due to electrolyte decomposition at low potentials, resulting from interactions between Li^+ ions, organic solvents, and electrons from the electrode. This layer plays a vital role in stabilizing the anode and enabling reversible lithium intercalation.

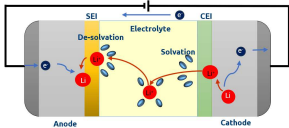


Fig. 1a Charge transfer process

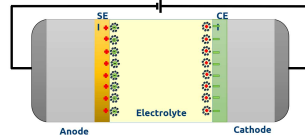


Fig. 1b Double-layer capacitance

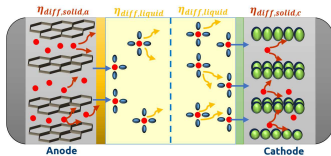


Fig. 1c Diffusion process

B. Lithium-ion Battery Degradation

Lithium-ion battery degradation is an intrinsic and complex process that occurs across various levels and modes, leading to a gradual decline in battery health and performance

over time. Key health indicators such as capacity fade, voltage fade, internal resistance, calendar aging, temperature sensitivity, and cycle life are impacted by degradation. The SOH is commonly used to quantify overall battery health. Degradation arises from a combination of time-dependent and usage-related factors, including extreme operating temperatures, overcharging and over-discharging, mechanical stress, current loads, and stoichiometry [26], as shown in Fig. 2.

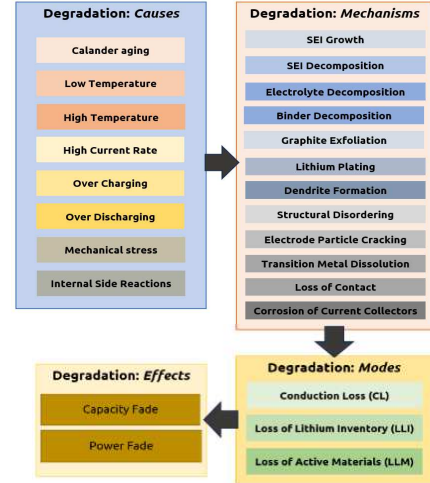


Fig. 2 Degradation of Lithium-ion batteries

Fast charging accelerates degradation; studies have shown that high charging currents promote active material loss (LAM), while high cutoff voltages contribute to lithium inventory loss (LLI) [27], [28]. Additionally, temperature plays a crucial role: at low temperatures, Li^+ diffusion and charge transfer resistance increase, promoting dendrite growth and thicker SEI layers [29]–[31]. High temperatures accelerate degradation through Arrhenius-driven reactions, including SEI and CEI breakdown, electrolyte combustion, and thermal runaway [32]. These degradation mechanisms are broadly categorized into three failure modes: loss of lithium inventory, loss of active material, and conduction loss, which collectively affect SOH and battery lifespan [33], [34].

C. Electrochemical Impedance Spectroscopy

Electrochemical Impedance Spectroscopy is a powerful, non-destructive technique used to analyze the electrical behavior of electrochemical systems. A sinusoidal voltage or current is applied across the system over a wide frequency range (usually from 1 MHz to 1 mHz), and the resulting response is measured [35]. EIS effectively deconstructs complex electrochemical processes by leveraging their distinct time constants ($\tau = RC$). These models processes such as electrolyte resistance, double-layer capacitance, charge transfer, and diffusion using equivalent circuit elements, offering clear insight into internal battery mechanisms. In LIBs, EIS aids in estimating ECM parameters for accurate modeling. Fast processes like charge transfer appear at high frequencies, while slower diffusion processes emerge at low frequencies, typically visualized on Nyquist plots as semicircles or line segments. Each LIB component is commonly represented by an electrical equivalent circuit shown in Fig. 3, while a lumped model (Fig. 4a) includes: R_s : ohmic resistance (electrode, electrolyte, separator, etc.); R_{ct} : charge transfer resistance at the electrode-electrolyte interface; C_{dl} : double-layer capacitance governing interfacial charge storage; Z_w : Warburg impedance representing lithium-

ion diffusion. Battery aging introduces an additional RC network (Fig. 4b), R_{SEI} and C_{SEI} , reflecting the resistance and capacitance of the SEI/electrolyte interface.

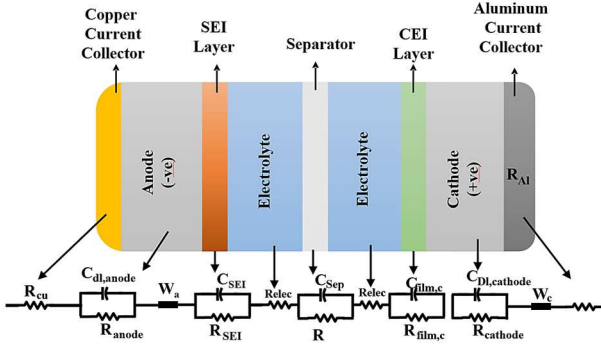


Fig. 3 Equivalent circuit for characteristic behavior of individual battery components

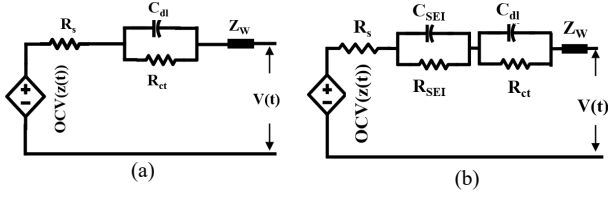


Fig 4 Lumped equivalent circuit model (a) 1-RC Model, (b) 2-RC Model

III. EXPERIMENTAL SETUP

A high-precision battery cycler with an EIS electrochemical tester and battery cycler, D&V BCT-150 cell tester, is used to test a cylindrical Samsung INR 21700 40T battery. This battery under test has a capacity of 4000 mAh and is made of LiNiMCoO₂ cathode with 4.2V maximum cutoff voltage and 2.5V minimum cutoff voltage, and a maximum continuous charge current of 6A (1.5C-rate). The test setup is shown in Fig. 5. Three different Samsung INR 21700-40T3 batteries were from the same batch are considered for the study and are labeled as cell-1(C1), cell-2 (C2) and cell-3 (C3). The specifications of the batteries are C1, pristine cell calendar aged for 8 months, C2, a cycle-aged cell for 330 charge-discharge cycles, and C3 a cycle-aged cell for 920 charge-discharge cycles. The battery details are listed in Table I.

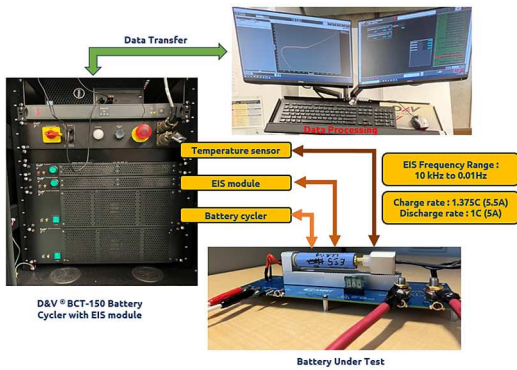


Fig. 5 Experimental Setup

TABLE I. Battery Parameters

Item	Specification
Battery Chemistry	Nickel Cobalt Aluminium (NCA)
Manufacturer and Model	Samsung 21700 40T3

Capacity	4000 mAh
Nominal Voltage	3.6 V
Nominal Charge Current	2 A (0.5C)
Rated Max. Charge Current	6 A

IV. DEEP NEURAL NETWORKS FOR EIS DATA EXTRACTION

Electrochemical impedance spectroscopy generates high-dimensional, frequency-dependent data that encodes valuable information about the physicochemical properties of electrochemical systems. However, conventional interpretation methods, relying on manual fitting of ECMs are often intensive, subjective, and prone to inconsistencies, particularly when multiple circuit topologies produce similar spectral features. To address these limitations, this work proposes a DNN-based framework to extract meaningful features from raw EIS data and automate the classification and interpretation process.

The approach formulates EIS analysis as a supervised learning task. Each EIS spectrum, initially represented as a complex-valued frequency response ($Z = Z' + jZ''$), is decomposed into a six-dimensional real-valued feature space comprising the real part, imaginary part, magnitude ($|Z|$), phase angle (θ), log-scaled frequency, and the derivative of phase concerning frequency ($d\theta/df$). This transformation enables the use of one-dimensional convolutional neural networks (1D-CNNs), which are well-suited for capturing local frequency patterns and trends across the spectral domain. The proposed model consists of a hierarchical stack of Conv1D layers with decreasing kernel sizes from 32 to 2. This is designed to capture both global and local spectral characteristics. These layers are followed by dropout and batch normalization to mitigate overfitting and improve generalization. A global average pooling layer condenses the learned features into a compact representation, which is then passed through dense layers to perform multi-class classification over four distinct ECM classes. To enhance robustness, the dataset is manipulated by adding Gaussian noise to more closely match the real world data collected with the experimental setup and the model is trained with early stopping and adaptive learning rate reduction. The model achieves high classification accuracy on both real and synthetic EIS data, even in the presence of noisy signals or overlapping circuit behaviors. Most importantly, the DNN architecture is not constrained by human-defined heuristics or curve-fitting algorithms. It learns directly from raw input, effectively mapping high-dimensional EIS spectra into a lower-dimensional embedding space that encodes discriminative circuit information. This capability enables the model to identify latent patterns that may be difficult for human interpreters or traditional machine learning models to detect [36], [37]. By automating the extraction and classification of EIS spectra, the proposed DNN framework accelerates the interpretive process, reduces human bias, and provides a scalable tool for real-time impedance monitoring in applications such as battery diagnostics, biosensors, and corrosion analysis.

Fig. 6 illustrates the architecture of the proposed DNN model used for the automated classification of EIS data. The model maps raw spectral features to their corresponding ECMs through a sequence of convolutional and dense layers. The input to the network consists of EIS data processed into three key real-valued features—imaginary part (Z''), phase

angle (θ), and magnitude ($|Z|$)—sampled across 100 frequency points. These are arranged as separate input channels, resulting in a tensor of shape (100×1) per feature, forming the input node structure. The input passes through two sequential 1D convolutional layers. The first layer contains 32 filters with a kernel size of 16, which captures broad frequency-dependent patterns. The second layer contains 64 filters with a kernel size of 8, enabling the model to learn more localized patterns. These convolutional layers help the model extract hierarchical features that represent the structure and behavior of the impedance spectrum.

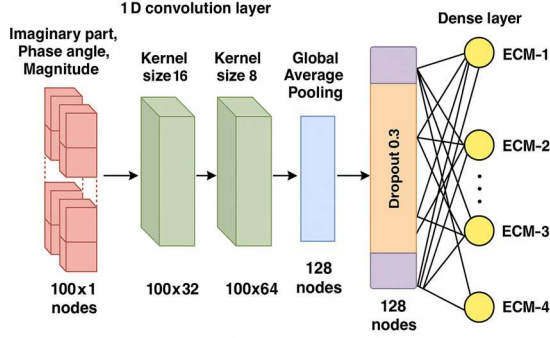


Fig. 6 Scheme of the neural network used for the EIS circuit classification

After a global average pooling layer compresses the 100-point sequences into a single 128-dimensional feature vector. This significantly reduces the number of parameters while preserving essential spectral information. To prevent overfitting and improve generalization, a dropout layer with a rate of 0.3 is applied, randomly deactivating 30% of the nodes during training. This encourages the model to learn robust, redundant representations that are less sensitive to noise or variability in the input. The resulting 128-dimensional feature vector is passed to a fully connected dense layer, which maps the extracted features to four output nodes—each corresponding to a specific ECM (ECM-1 to ECM-4). A SoftMax activation function is applied at the output layer to return class probabilities, enabling reliable classification based on the highest-likelihood match. The detailed model parameters and training configurations are listed in Table II.

TABLE II Model Parameters and Training Configurations

Layer	Input
Details Input Shape	(100, 6)
Conv1D-1	64 filters, kernel size = 16, activation = ReLU
Conv1D-2	128 filters, kernel size = 8, activation = ReLU
Dropout	Rate = 0.3 (to reduce overfitting)
Normalization	BatchNormalization
Pooling	GlobalAvgPooling1D
Dense	256 units, activation = ReLU
Output (Dense)	4 units, activation = Softmax (for 4 ECM classes)

V. DNN PREDICTED ECM AND NYQUIST PLOTS

Once the DNN model was trained and validated, it was applied to classify unseen EIS spectra based on their underlying ECMs. The DNN output applied SoftMax across four ECM classes, highlighting the highest valued circuit configuration associated with the input spectrum. Each ECM, as shown in Fig. 7, represents a simplified electrical analog that emulates electrochemical behavior at the electrode–

electrolyte interface. The Nyquist plots corresponding to each predicted ECM class were analyzed to visually validate the classification and assess the shape and key features of the impedance spectra.

A. ECM-1 (Single semicircle – Randles-type circuit):

This model typically consists of a series resistance (R_s) in series with a parallel combination of a resistor (R_l) and a constant phase element (Q_l), as shown in Fig. 7a. The Nyquist plot of ECM-1, as shown in Fig. 8a exhibits a single, well-defined semicircular arc centered on the real axis, indicating a single time constant associated with interfacial charge transfer. The DNN accurately classified these spectra, and the plots matched the canonical Randles-type behavior.

B. ECM-2 (Two semicircles – Dual RC branches):

This circuit contains R_s in series with two $R \parallel Q$ branches, capturing both high- and low-frequency processes, as shown in Fig. 7b. The Nyquist plots of ECM-2, as shown in Fig. 8b, typically show two distinguishable semicircles, with the high-frequency arc corresponding to double-layer charging and the low-frequency arc representing slower diffusion-related processes. The DNN showed high confidence in identifying ECM-2, especially when the two arcs were well-separated.

C. ECM-3 (Single arc + Warburg tail – Diffusion-dominated circuit):

ECM-3 includes a Warburg diffusion element (Z_w) in parallel with a capacitor, following the $R_s + [R_l + Z_w \parallel Q_l]$ topology, as shown in Fig. 7c. The corresponding Nyquist plots, as shown in Fig. 8c show a semicircular arc followed by a 45° linear tail at low frequencies, which is indicative of semi-infinite diffusion. The DNN correctly identified this signature and maintained consistent accuracy even with varying Warburg coefficients.

D. ECM-4 (Combined diffusion and charge-transfer dynamics):

This complex circuit includes R_s in series with multiple RC elements and a Warburg component, resulting in spectra with a combination of two semicircles and a diffusion tail, as shown in Fig. 7d. The Nyquist plots for ECM-4, as shown in Fig. 8d, are more intricate, often with overlapping arcs and angled low-frequency regions. Despite this complexity, the DNN model successfully captured the distinguishing patterns and classified these spectra with high accuracy. In each case, the Nyquist plots of the DNN-predicted class were found to be in strong agreement with theoretical expectations for the corresponding circuit. This correlation further supports the reliability of the model in both classification accuracy and interpretability of its predictions.

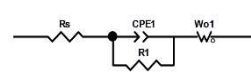


Fig. 7a ECM-1 Circuit

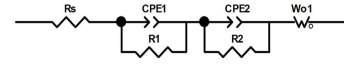


Fig. 7b ECM-2 Circuit

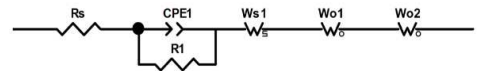


Fig. 7c ECM-3 Circuit

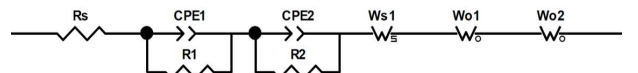


Fig. 7d ECM-4 Circuit

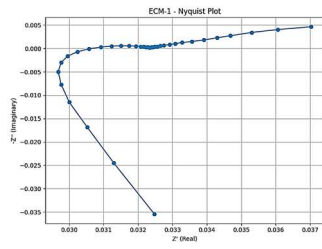


Fig. 8a Nyquist plot of ECM-1

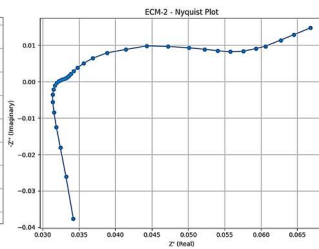


Fig. 8b Nyquist plot of ECM-2

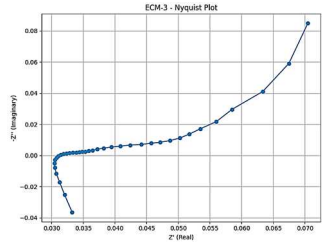


Fig. 8c Nyquist plot of ECM-3

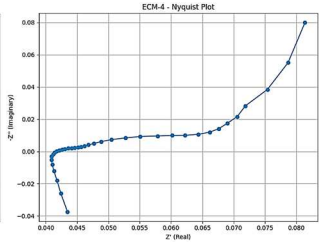


Fig. 8d Nyquist plot of ECM-4

VI. DNN TRAINING AND EVALUATION OF EIS MODELS

The performance of the proposed DNN model was evaluated on the test dataset after training on augmented and normalized EIS spectra. The model achieved a high classification accuracy of 95.65% on the test set, indicating excellent generalization capability across all four ECMs. As shown in Table III and Fig. 9, the model maintained stable learning behavior throughout training, with no signs of overfitting as evidenced by the steady and consistent decline in validation loss over 20 epochs.

TABLE III Results Obtained

Metric	Value
Test Accuracy	95.65%
Best F1-scores	ECM-0 to ECM-3 all above 0.88+
Model Stability	No signs of overfitting (Val loss dropped steadily)
Epochs	20 (fast convergence)

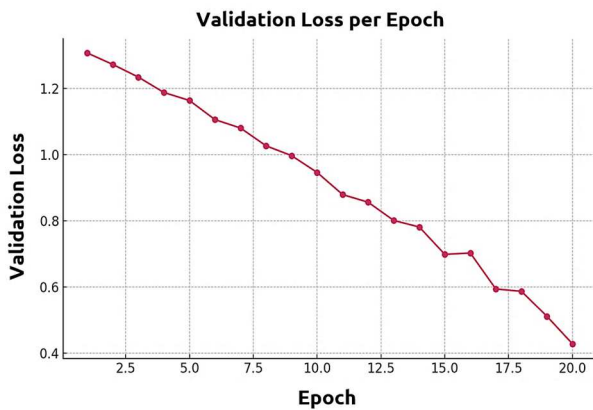


Fig. 9 Validation loss per Epoch

The F1-scores for all classes (ECM-0 to ECM-3) were above 0.88, confirming both high precision and recall across circuit classes. The confusion matrix, as shown in Fig. 10, further supports this result, where most predictions align along the diagonal, indicating correct classification. Specifically, ECM-1, ECM-2, and ECM-4 achieved perfect or near-perfect classification, while ECM-3 experienced a single misclassification, likely due to partial overlap in spectral

features with ECM-4. The bar plot, as shown in Fig. 11, compares the actual versus predicted output across all ECM classes and demonstrates that the predicted class distribution closely matches the ground truth. Additionally, the validation loss graph shows consistent improvement over time, reaching convergence within 20 epochs. This fast convergence indicates that the model architecture and hyperparameter settings were well-tuned for the task. These results collectively validate the robustness and accuracy of the DNN model in classifying EIS spectra and highlight its potential for rapid, automated interpretation of complex electrochemical data.

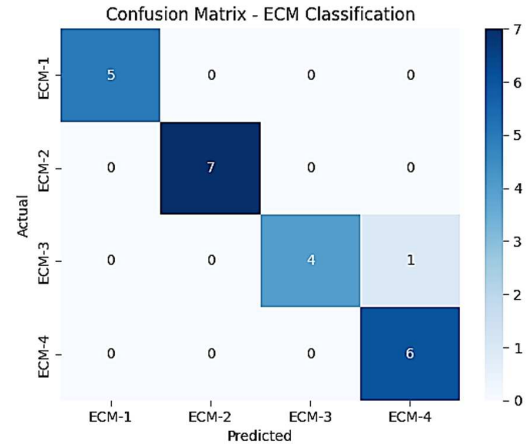


Fig. 10 Confusion Matrix

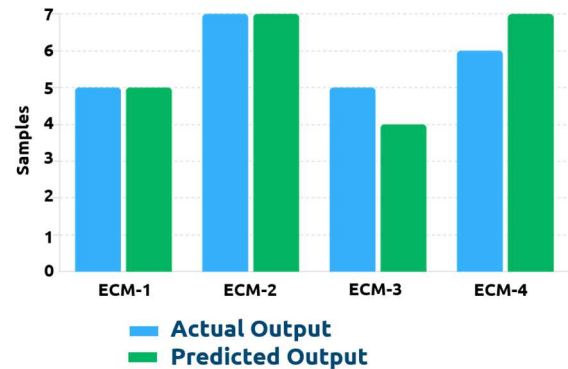


Fig. 11 Actual and Predicted Output

VII. CONCLUSIONS

In this study, the authors present a deep learning-based framework for the automated classification of ECMs from EIS data to improve lithium-ion battery diagnostics. By employing a 1D-CNN, the proposed model learns to extract and interpret frequency-domain features, enabling accurate classification of EIS spectra into four distinct ECM categories with a test accuracy of 95.65%.

The model demonstrates strong performance in identifying key electrochemical processes such as charge transfer resistance, double-layer capacitance, and diffusion dynamics. The predicted ECMs show high consistency with the theoretical characteristics observed in corresponding Nyquist plots, validating the interpretability and robustness of the approach. Furthermore, the deep neural network generalizes well across both synthetic and experimentally acquired datasets, even in the presence of noise or overlapping circuit features. This highlights the potential of data-driven methods in automating EIS analysis, reducing subjectivity,

and accelerating battery health diagnostics. The authors suggest that future extensions may involve adapting the model to accommodate more complex circuit topologies and integrating it into real-time BMS for advanced health monitoring and predictive maintenance.

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